

CubeSat Two-Body Orbit Propagator and Deployment Sensitivity Study

A six-state orbit propagator built from the two-body equations of motion in Simulink, verified against closed-form orbital mechanics, and used to quantify how a fraction-of-a-percent tangential velocity error at CubeSat deployment propagates into perigee, apogee, eccentricity, and orbital period.

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Problem statement

A CubeSat dispenser cannot deploy a spacecraft at exactly the intended velocity. Spring-based deployers have tolerances, and the host vehicle itself has insertion error. This project asks a specific question:

If a CubeSat intended for a 400 km circular orbit leaves the dispenser with a small tangential velocity error, how much does the resulting orbit differ from the target?

The answer is used to build intuition for why Δv budgeting dominates mission planning: a velocity error applied at one point on an orbit produces almost no change *there*, and its full effect on the opposite side.

Model

The propagator integrates the two-body equations of motion in an Earth-centered inertial frame:

$$\dot{\mathbf{r}} = \mathbf{v}$$

$$\dot{\mathbf{v}} = -\mu \mathbf{r} / \|\mathbf{r}\|^3$$

where

Symbol	Meaning	Value / units
μ	Earth standard gravitational parameter (GM)	$3.986004418 \times 10^{14} \text{ m}^3/\text{s}^2$
R_e	Earth equatorial radius (WGS-84)	6378.137 km
$\mathbf{r}$	Position vector, Earth center to spacecraft	m
$\mathbf{v}$	Inertial velocity vector	m/s
$dvFrac$	Fractional tangential velocity error at deployment	dimensionless

The Simulink model is a single closed loop: one Integrator carries all six states $[\mathbf{x} \ \mathbf{y} \ \mathbf{z} \ v_x \ v_y \ v_z]^T$, a MATLAB Function block computes the state derivative, and the derivative feeds back to the Integrator input. A To Workspace block logs the state history.

Solver configuration

Setting	Value
Solver type	Variable-step
Solver	ode45
Stop time	t_{Final} (6000 s, ≈ 1.08 orbits)
Max step size	10 s
Relative tolerance	1e-10
Absolute tolerance	1e-6

The tightened relative tolerance matters. Position states are on the order of 7×10^6 m, so Simulink's default `RelTol` of 1e-3 admits kilometres of numerical drift per orbit and makes the energy conservation check meaningless.

Verification

The two-body problem conserves specific orbital energy and specific angular momentum exactly, so any drift in either is pure integration error. Three independent checks were run on the nominal case:

Check	Result	Interpretation
Orbital period vs. closed-form $2\pi\sqrt{a^3/\mu}$	92.56 min	Matches analytic value
Specific energy drift over one orbit	$6.957 \times 10^{-11} \%$	Negligible
Specific angular momentum drift	$3.477 \times 10^{-11} \%$	Negligible
Eccentricity, nominal circular case	0.00000	Circular orbit stays circular

A vis-viva cross-check ($v^2 = \mu(2/r - 1/a)$) is included in `analyze_orbit.m` and agrees with the propagated speed to within numerical precision.

Results

Circular velocity at 400 km altitude is **7668.6 m/s**, so a 0.25 % error is approximately **19 m/s**.

Case	Δv	Perigee (km)	Apogee (km)	Eccentricity	Period (min)
Nominal	0 %	400.0	400.0	0.00000	92.56
Fast insertion	+0.25 %	400.0	468.2	0.00501	93.26
Slow insertion	-0.25 %	332.6	400.0	0.00499	91.87
Fast insertion	+0.50 %	400.0	537.3	0.01003	93.97
Slow insertion	-0.50 %	266.1	400.0	0.00998	91.19

Case	Δv %	Perigee (km)	Apogee (km)	Eccentricity	Period (min)
Fast insertion	+1.0 %	400.0	678.1	0.02010	95.42
Slow insertion	-1.0 %	135.5	400.0	0.01990	89.87

Limitations

- **Sensitivity is roughly linear** in Δv % for errors of this magnitude, since the orbit remains near-circular and the perturbation is small compared to circular velocity.
- **The slow case has an operational consequence this model does not capture.** A depressed perigee sweeps denser atmosphere once per revolution, accelerating orbital decay and shortening mission lifetime — a real effect at 400 km that requires a drag model to quantify.

Figures

